

1 Latent Relational Source Generation

Inputs

- Domain
- Attribute count
- Attribute cardinality
- Past table names

Schema
generation

Table
generation

Latent relational source

Company	Facility	...	Value
X	Atlanta	...	42.1
Y	Busan	...	31.4
Z	London	...	19.2
...	...	...	...

SQL Template
Filling

SQL Query
Execution

Aggregation
Execution

$$g \in \{\text{SUM, AVG, MAX, MIN}\}$$
$$y = g(a_1, \dots, a_3)$$

SQLs & Answers

Parallel

one scalar per table, then aggregate

```
-- T1:SELECT value FROM R WHERE company='X'AND branch='UK'→ a1
-- T2:SELECT value FROM R WHERE company='Y'AND branch='UK'→ a2
-- T3:SELECT value FROM R WHERE company='Z'AND branch='UK'→ a3
```

answer
parallel

Sequential

output of step i becomes lookup key of step i+1

```
-- T1:SELECT company FROM CEOs WHERE ceo='John Doe'→ b1
-- T2:SELECT facility FROM Assign WHERE company=b1→ b2
-- T3:SELECT value FROM R&D WHERE facility=b2
      AND sector IN ('AI', 'Finance') → {a1, a2}
```

answer
seq.

Latent Relational Source + Query Provenance Mask

3 Rendering Non-Relational Views

Transformations

Layout edits

- Pivot (hierarchical headers)
- Column merging
- Blank row/column insertion
- Attribute name change

Cell edits

- Typographical noise
- Footnote / text insertion
- Null / N.A. insertion

Numerical edits

- Unit-of-measurement change

Parallel views : each view contributes one independent scalar

Company X			Company Z		
2023 indicators / employee	Branch location		Weekly usage per employee	Rbnach citise	
	UK	US		Lodnon (UK)	Tokoy (Jaapn)
Weekly water usage 2022	5.89 gallons	7.16 gallons	Waetr	2022	0.022m ³
Weekly water usage 2023	5.60 gallons	--		2023	0.024m ³

Sequential views : each lookup key parameterizes the next view

CEO names			R&D expenses		
Year	Compaines		Costs (2023)	Investments (millions)	
	X	Y		AI Research	Finance Research
2022	John Doe	Lee Seo-yeon	Net cost	Atlanta Lab	42.1
2023	Jnae Mya	n.a.		Seoul Lab	53.8 - as written in [1]

4 Question Verbalization & Verification

Question
Generation

Question
Verification

parallel NL question

sequential NL question

Tables

parallel
non-relational
tables

sequential
non-relational
tables

Natural Language Questions

What is the average water consumption in liters of the top 3 S&P 500 companies in 2023, considering only UK branches?

What is the average 2023 investment of the company where John Doe is CEO? Restrict to AI and the financial sector.

Answers

22.23

40.5